Stockholm University

Department of Meteorology

Modelling of Large-Scale Circulation in Atmosphere and Ocean (MO8004)

Rossby Waves
Lab 3

May 21, 2026

Author: Fredrik Bergelv (fredrik.bergelv@live.se)

Supervisor: Verena Jung

Questions

Theory

1. State the necessary assumptions that are needed to study Rossby waves.
2. Write down the dispersion relation of Rossby waves, the equation for the group velocity, and the phase speed. Discuss the dispersion of Rossby waves.
3. Discuss in your own words the term dispersion.
4. From the equations in Question 2, discuss the role of the wave number k in relation to the Rossby radius R. Which cases can be distinguished?
5. State the equation for $\partial h/\partial t$ with variable depth

$$D = D_0 + \alpha y$$

   and indicate the two additional terms that need to be implemented.
6. State the relation where α cancels the β-effect.
7. Why do we have to assume reduced gravity in this laboratory?

Experiments

Set up the model as described in the laboratory description for Laboratory 3B and follow the instructions for the experiments.

2.1 The role of wave number

1. List six values of kR and the corresponding L_w that sample the characteristic shape of the zonal Rossby wave phase speed c_x and group speed c_{gx} relations, assuming

$$L_w = \frac{1.3\sqrt{g'H}}{kR\sqrt{5}\, f_0}.$$

   Explain your choice.
2. Plot the theoretical zonal phase speed c_x and group speed c_{gx} as a function of kR, and highlight the values corresponding to your choices in Question 1.
3. Choose three kR values representing the characteristic Rossby wave regimes and provide Hovmöller diagrams at the center of the y-axis. Estimate the numerical zonal phase and group velocities and plot them as lines in the diagrams. Add the theoretical zonal phase and group speeds as lines in the Hovmöller diagrams.
4. Add the numerical estimates of your three characteristic waves into the plot from Question 2. (It is acceptable to combine Questions 2 and 4 into one final figure.)

2.2 The role of topography

1. Fix β and choose a value of kR for experiments with varying α. Plot Hovmöller diagrams of at least three representative waves on the α-plane. Add the theoretical zonal phase and group speeds to the plots and explain your parameter choices.
2. Plot the waves shown in Question 1 as xy cross-sections at an appropriate time.
3. For two of your α-value experiments (excluding the cancelling case), plot the theoretical c_g and c_x as a function of kR and include the numerical estimates from the experiments. Compare the results with the plot from Question 2.1.4.

1 Theory

i) In order to study topographic Rossby waves one needs four critical assumptions;

1. Assumption that the potential vorticity, defined as

$$\frac{D}{Dt}(P) = 0 \text{ where } P = \frac{\xi + f}{h}, \tag{1}$$

   is conserved. Above $\xi = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)$ is the relative vorticity, f is the Coriolis parameter and h is the fluid depth. Conserved potential vorticity means that we neglect dissipation from e.g. friction.

2. The beta-plane approximation, where we assume Coriolis changes with latitude as $f \approx f_0 + \beta y$. Here β is the linearized Coriolis parameter around a point f_0. We also need that $\frac{\beta y}{f} \ll 1$, which is generally true. The value of the beta plane can be found as $\beta = 2\Omega\cos(\phi_0)/a$, where Ω is the angular rotation rate of the Earth and a is the radius of the Earth.

3. A slowly varying bottom depth as $H(x, y) = H_0 + \alpha y$, where $\frac{\alpha y}{H_0} \ll 1$.

4. Quasi-geostrophy, i.e. that we are almost in geostrophy. This is the same as stating that the Rossby number $R = \frac{U}{fL}$ is small ($R \ll 1$). Here U is the characteristic flow speed and L is the characteristic domain size.

ii) Using the assumptions presented above, the dispersion relation for a Rossby wave with a meridionally varying topography can be derived as

$$\omega = -k\frac{\beta - \frac{f_0}{H_0}\alpha}{k^2 + l^2 + \frac{1}{R_d^2}}, \tag{2}$$

where k is the zonal wave number, l is the meridional wave number and R_d is the Rossby radius of deformation. Assuming no meridional waves ($l = 0$), one can derive the zonal phase speed as

$$c_{p_x} = \frac{\omega}{k} = -\frac{\beta - \frac{f_0}{H_0}\alpha}{k^2 + \frac{1}{R_d^2}}, \tag{3}$$

and the zonal group velocity as

$$c_{g_x} = \frac{\partial\omega}{\partial k} = \left(\beta - \frac{f_0}{H_0}\alpha\right)\frac{k^2 - \frac{1}{R_d^2}}{\left(k^2 + \frac{1}{R_d^2}\right)^2}. \tag{4}$$

Above one can observe a dispersive type of wave where the phase and group speeds are generally not the same. The dispersion is also dependant on the tilt, mean depth and Coriolis parameter as $-\frac{f_0}{H_0}\alpha$. It is also dependent on the beta-effect as β. The wave-numbers also appear beside the Rossby radius of deformation, meaning that this is the characteristic length-scale of the system.

iii) Dispersion is simply the case when the phase speed is different from the group velocity. Another way of stating the same definition is that the wave is dispersive if there is a wave-number dependence in the phase speed, i.e. that different waves have different phase speeds. This is the same since both cases are true if the dispersion equation is not linear in k. Thus observing Equation 2 one can see that this solution is dispersive, since the solution is not linear in k. The dispersive solution can also be observed by observing phase and group speeds not being the same. However, if one observes the long-wave limit $k \ll \frac{1}{R_d}$, then we see that Equation 2 becomes linear in k. Thus the long-wave limit is non-dispersive.

iv) Varying the wave number k can be done to characterize the length scale, i.e. the Rossby radius of deformation R_d. One can from this distinguish different cases.

1. **The long-wave limit:** The long-wave limit is when $k \ll \frac{1}{R_d}$, i.e. we have long wavelengths. Observing Equation 2 one can see that in this limit, the denominator k^2 goes to zero. The dispersion relation thus becomes linear in k, and the phase and group speeds become the same

$$c_{p_x} = c_{g_x} = -\left(\beta - \frac{f_0}{H_0}\alpha\right) R_d^2. \tag{5}$$

   One can furthermore note that if the topographic effect is less than the beta-effect, then we get non-dispersive movement towards the West.

2. **The short-wave limit:** The short-wave limit is when $k \gg \frac{1}{R_d}$, i.e. we have very short wavelengths. Observing Equation 2 one can see that the $\frac{1}{R_d^2}$ becomes irrelevant. One can thus obtain the phase and group speeds as

$$c_{p_x} = -\frac{\beta - \frac{f_0}{H_0}\alpha}{k^2} \quad c_{g_x} = \frac{\beta - \frac{f_0}{H_0}\alpha}{k^2}. \tag{6}$$

   Here one can see that the wave-packet will propagate to the East, while the phase speeds propagate to the West. However, since we assumed that the wave number is really large we can see that both the phase and group speeds go to zero, i.e. $c_{p_x} \to 0$ and $c_{g_x} \to 0$ when $k \to \infty$.

3. **The eddy limit:** The final case is when $k = \frac{1}{R_d}$, or $kR_d = 1$. Here we can rewrite the phase and group velocities as

$$c_{p_x} = -\left(\beta - \frac{f_0}{H_0}\alpha\right)\frac{R_d^2}{2} \quad c_{g_x} = 0. \tag{7}$$

   Here one can see that the group velocity becomes zero, i.e. that we have a stationary wave-packet. However, the phase speed is not zero, meaning that the wave-packet will move internally.

v) The normal convergence equation in the shallow water equations is

$$\frac{\partial \eta}{\partial t} + \frac{\partial (uh)}{\partial x} + \frac{\partial (vh)}{\partial y} = 0. \tag{8}$$

Normally we use that $h = H_0 + \eta$, where $\eta \ll H_0$. But now we instead use a varying topography with $h = H_0 + \alpha y + \eta$, where $\eta \ll H_0$ and $\alpha y \ll H_0$. Using this on Equation 8 we obtain a new convergence equation as

$$\implies \frac{\partial \eta}{\partial t} + H_0\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + \underbrace{\alpha y\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + \alpha v}_{\text{new topographic terms}} = 0. \tag{9}$$

Thus when solving for $\frac{\partial \eta}{\partial t}$ we get three new terms, $-\alpha y\frac{\partial u}{\partial x}$, $-\alpha y\frac{\partial v}{\partial y}$ and $-\alpha v$ on the right-hand side.

vi) Observing Equation 8–4 one can see that the beta-effect is cancelled when

$$\beta = \frac{f_0}{H_0}\alpha. \tag{10}$$

If one has a beta value $2 \times 10^{-11}\,\mathrm{s^{-1}\,m^{-1}}$, a Coriolis parameter of $1 \times 10^{-4}\,\mathrm{s^{-1}}$ and a mean depth of $4000\,\mathrm{m}$ one obtains an α-value of $8 \times 10^{-4}\,\mathrm{m^{-1}}$.

vii) Reduced gravity is an essential tool to get more physically realistic solutions. Previously, the Rossby radius of deformation was here defined as

$$R_d \equiv \frac{\sqrt{g\prime H}}{f_0}.$$

If one was to compute the Rossby radius of deformation with some common parameters such as $H = 4000\,\mathrm{m}$, $f_0 = 1 \times 10^{-4}\,\mathrm{s}^{-1}$ and a normal gravity of $g = 9.82\,\mathrm{m\,s^{-2}}$, one would obtain a Rossby radius of almost 2000 km, which is more than the distance from Stockholm to Istanbul! Thus this is a highly unrealistic value. If one instead uses a reduced gravity of $g\prime = 0.0981\,\mathrm{m\,s^{-2}}$ one gets a radius of 200 km, which is far more realistic.

Furthermore, most Rossby waves in the oceans and the atmosphere are baroclinic and not barotropic. This is because the temperature/density varies with depth/altitude. Thus it is more realistic to consider a baroclinic scenario, rather than a barotropic one.

2 Experiments

2.1 The role of wave number

i) We know that the width of the initial Gaussian, L_w is related to the wave number and Rossby radius as

$$L_w = 1.3\frac{\pi}{kR_d}\frac{1}{\sqrt{5}}\frac{\sqrt{g\prime H}}{f_0}.$$

Using this Table 1 could be created.

kR_d	L_w (km)	% of domain
0.50	723.61	10.3372
1.00	361.80	5.1686
1.73	209.13	2.9876
3.00	120.60	1.7229
5.00	72.36	1.0337
7.00	51.69	0.7384

Table 1: This table displays the kR_d-points which were selected for analysis, their corresponding L_w, and that L_w expressed as a percentage of the domain length (7×10^6 m).

The points in Table 1 were selected to sample as many points from the theoretical shape (which can be seen in Figure 2). Since the dispersion relation is symmetric in the plus/minus domain, only positive points were chosen (since negative points would just give the antisymmetric speeds). The especially important points [0.5, 1, 1.73, 3, 5, 7], were argued from Equation 2. The point $kR_d = 0.5$ was chosen since this corresponds to the long-wave limit, where we expect to see less dispersive waves which move more slowly. The point $k = \frac{1}{R}$ (i.e. $kR_d = 1$) was chosen since at this point we expect to see a stationary wave packet, as seen in Equation 4 where this gives $c_{g_x} = 0$. The point $kR_d = 1.73$ was chosen since this is the point where we would observe the strongest eastward group velocity. The points [3, 5, 7] were chosen to examine the short-wave limit where we expect to see a convergence towards stationary, non propagating waves.

iii) A Hovmöller diagram showcasing the wave for the different kR_d-points from Table 1 can be seen in Figure 1. There one can see the group speed as the direction between the peaks and valleys. The phase speed can be seen as the temporal line of peaks from the initial position. One can observe good agreement from the values where $kR_d \geq 1$. Especially good results can be observed when $kR_d = 1$, where the theoretical and numerical phase/group speeds really align. The point $kR_d = 0.5$ clearly gave the worst result, where a large difference between the numerical and theoretical speeds can be observed.

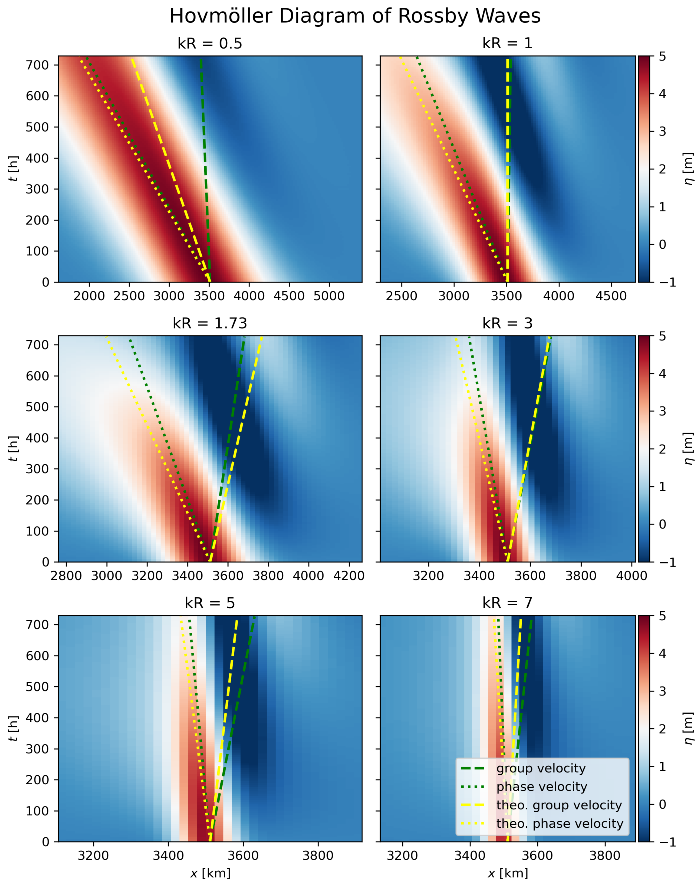

Figure 1: This figure showcases the phase and group propagation for the different kR_d-points from Table 1.

ii, iv) One can observe the differences between the numerical and theoretical phase/group speeds together with the theoretical curves in Figure 2. There one can observe more clearly the differences seen in Figure 1. Again one can observe good agreement when $kR_d \geq 1$ and bad agreement when $kR_d = 0.5$. The reason for this long-wave mismatch can be described by the nature of long numerical Rossby waves having a harder time to fit the domain. Since the domain is suppose to act as an infinite ocean, having a larger Rossby wave-length means that the finite domain may become a source of error.

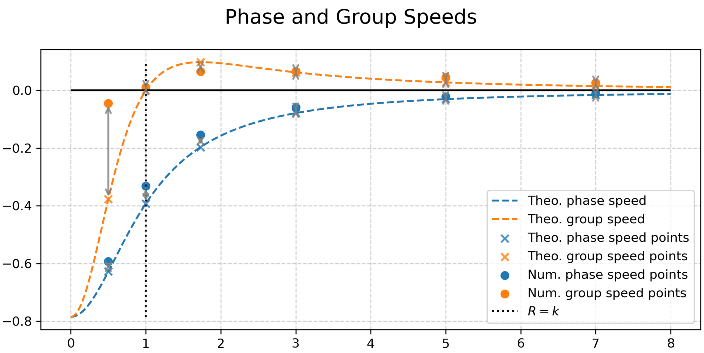

Figure 2: This figure shows the theoretical phase and group speeds together with the theoretical and numerical speeds for the kR_d-points from Table 1.

2.2 The role of topography

i) Choosing $kR_d = 1.73$ and introducing topography as $H(y) = \alpha y + H_0$ gives the result seen in Figure 3. For the case when $\alpha = 0$ one can see an East-moving wave-packet and Westward phase, as expected from the theory. When α was increased to $2.29 \times 10^{-4}\,\mathrm{m}^{-1}$ and $4.58 \times 10^{-4}\,\mathrm{m}^{-1}$ one can observe less propagation on the same timespan, still with Westward phase and Eastward group velocity. However, when the topography tilts in the opposing way, $\alpha = -2.29 \times 10^{-4}\,\mathrm{m}^{-1}$, we see an even faster moving wave compared to when there was no topography. This makes sense, since a negative tilt would only increase the "observed beta-effect". When $\alpha = 8 \times 10^{-4}\,\mathrm{m}^{-1}$ nothing moves, as expected from the theory. More interestingly, one can observe a Westward group velocity and Eastward phase when the α-value was set to be above the cancelling α-value.

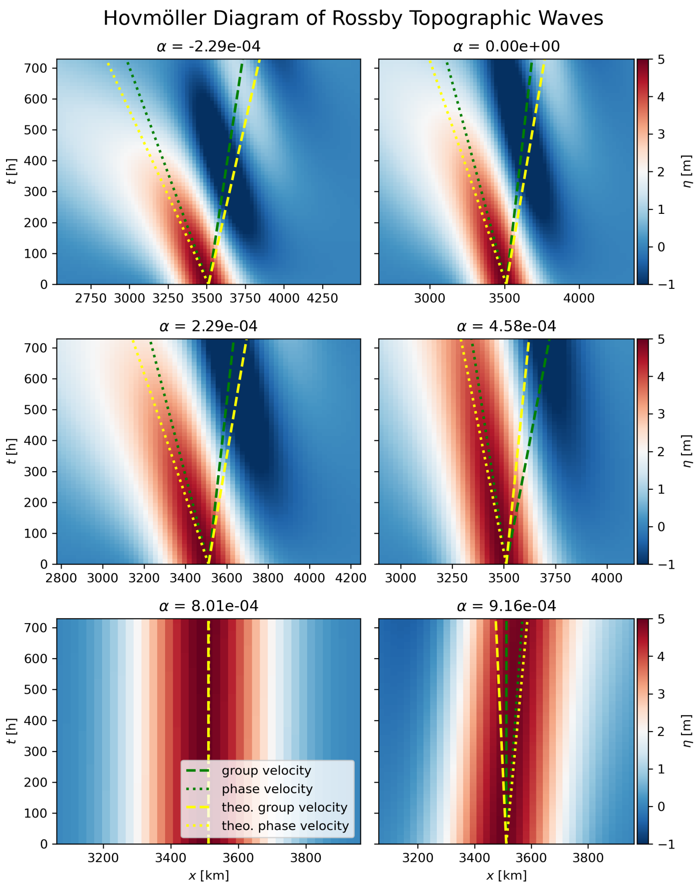

Figure 3: This figure shows the topographic Rossby waves for $kR_d = 1.73$ as Hovmöller diagrams.

ii) The same result observed in Figure 3 can be seen for the entire domain in Figure 4. There one can see results for the initial condition, 10 days, 20 days, and 30 days. An interesting observation is that even though setting $\alpha = 8 \times 10^{-4}\,\mathrm{m}^{-1}$, the edges of the initial perturbation can still be seen to move due to the beta-effect.

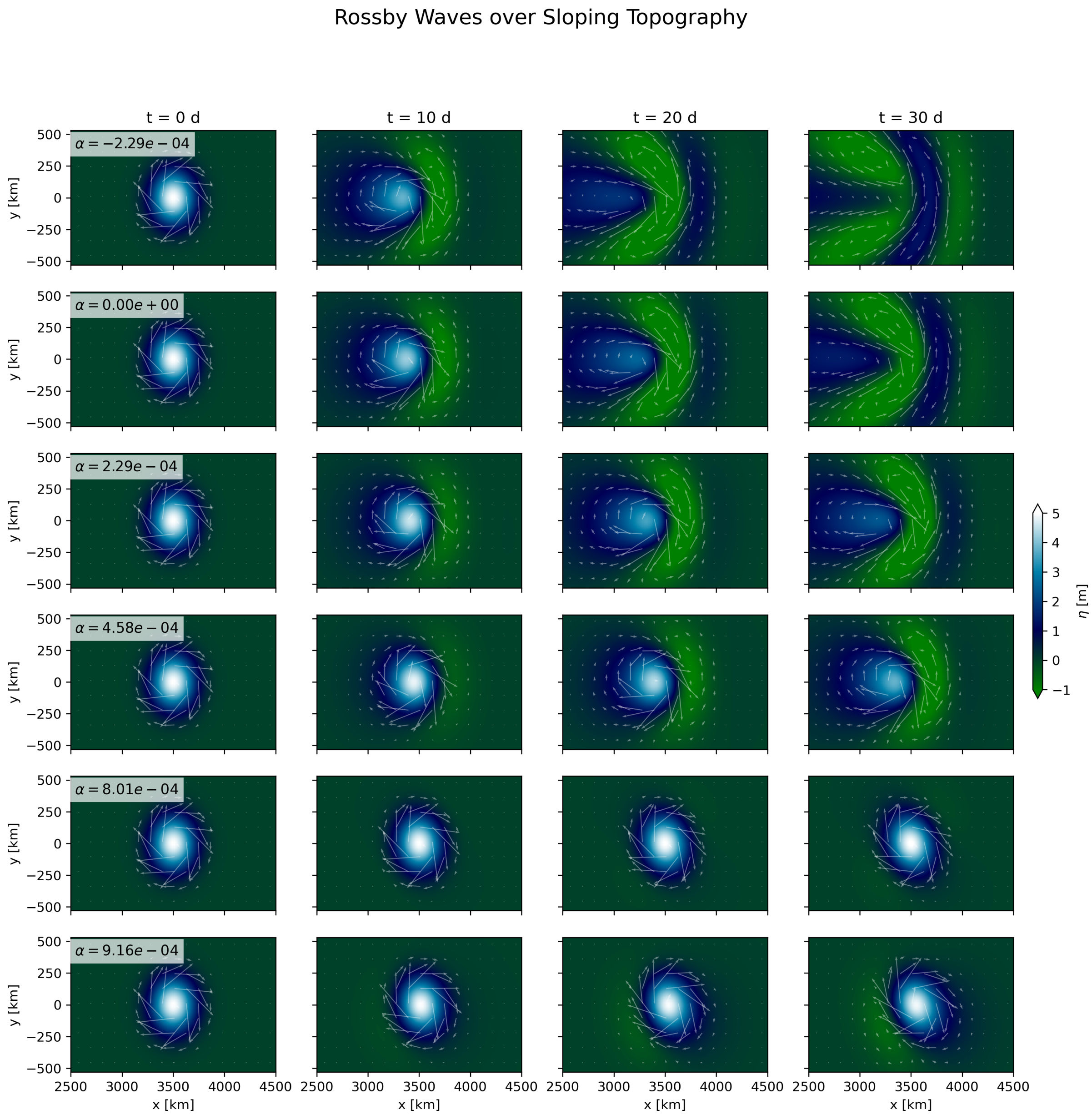

Figure 4: This figure shows the contour plots of the Rossby waves in the xy-plane.

iii) In Figure 5 one can observe the phase and group speeds like in question 2.iv. Here one can see that a negative α-value made stronger velocities, with a similar accuracy to the non-topographic curve. The middle plot with $\alpha = 4.58 \times 10^{-4}\,\mathrm{m}^{-1}$ showed damped speeds compared to the non-topographic curve, again with similar accuracy. The α-value with a higher value than the cancelling α-value showed opposing speeds compared to the others. This is because the topography is now stronger than the beta-effect, which makes the Rossby wave propagate in the opposing direction.

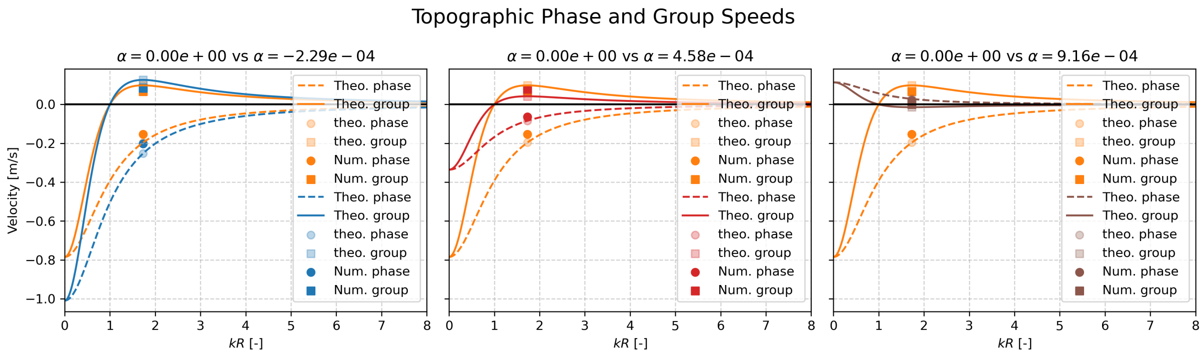

Figure 5: This figure showcases the phase and group velocities for a few different α-values, compared to $\alpha = 0$. The $\alpha = 0$-line is always the orange one in all subfigures.

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`